

is fed to pin 2 and pin 7 of IC3, which drives the DC motor in one direction. This enables the car to move forward. IC5 is wired as an astable multivibrator to generate a 38kHz pulse.

The 38kHz pulses are amplified by transistor T1 to drive five IR LEDs. The IR sensors and LEDs are arranged in such a way that the output of the sensors is high where there is no obstacle near the car. But if there is any obstacle, the transmitted IR signals from the IR LED are reflected back from the obstacle that comes in the way. This makes the sensor output go low. This low output signal is fed to port 0 of the microcontroller. The output of sensor

Q1 is connected to P0.0, Q2 to P0.1 and Q3 to port P0.2.

As soon as the output of the left sensor Q1 is pulled low to P0.0=0, the controller alters the spin of the DC motor, makes port P3=0 and stops the DC motor, thereby stopping the car for a moment. Then it drives port 1 in such a way that the stepper motor is rotated to the right, and the front steering wheels are also turned right, while the car continues to move forward. After a few seconds, the controller drives the stepper motor in the reverse direction (turning it left), which brings the steering wheels to the straight position and moves the car forward. Similar is the case when an obstacle is detected at the right sensor Q2.

In case there is an obstacle in the front, the centre sensor Q3 sends a low signal to port pin P0.2. This makes port P3=0 and the car stops. The steering wheels are turned left making port P3.0=0

and P3.1=1, which moves the car in backward direction with the steering still turned. After a few seconds, the DC motor is stopped by making port P3=0. The steering is set in the straight position and the car moves forward by making the port P3.0=1 and P3.1=0.

Thus, when the port P3.0=0 and P3.1=1, the car moves in the reverse direction; when P3.0=1 and P3.0=0, the car moves in the forward direction and when P3.0=0 and P3.1=0, the car stops.

Construction

Fig. 3 shows the plan view of the mechanical model of the car. The rear wheel is driven by a belt, pulley and a DC motor. The pulley and drive belt for the rear wheels are taken from the HP Business Inkjet 1000 ink tank pump assembly.

If the polarity of the DC is reversed, the motor spins in the opposite direction. The same concept is used for driving the DC motor. The front wheels are mounted on a fibre gear which can be rotated in both directions using a gear and a stepper motor. The gears are taken from an old HP DeskJet 3325 paper-feed printer assembly and the stepper motor for steering control is taken from an old Epson Inkjet 460 printer assembly.

In order to avoid collision or friction between the main car chassis and the wheels, enough space has been provided between them for their free rotation. The front PCB is mounted at a height to avoid collision with front wheels while turning. Two gears are used for reducing the speed when the car is running on a rough surface. The main chassis is made of a thick fibre board.

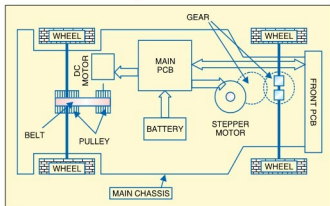


Fig. 3: Plan view of the mechanical model of the RoboCar

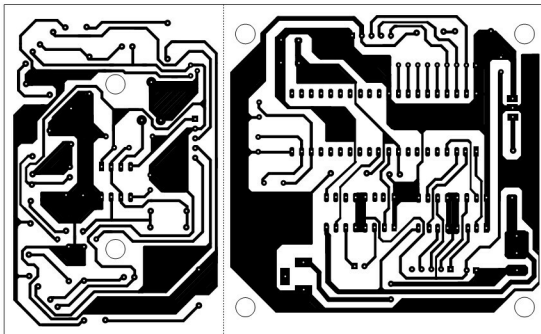


Fig. 4: A single-side, actual-size PCB layout for the AT89C52-based RoboCar